

# A Global Misalignment Index for AMR R&D Funding: Quantifying Where Investment Falls Short of Burden, and Where the Pipeline Compounds the Gap

## Abstract

Global antimicrobial resistance (AMR) R&D draws an estimated USD 18.9 billion across 18,853 projects, yet whether this investment aligns with the disease burden it addresses has not been quantified at the pathogen level. We construct a Global Misalignment Index (GMI) for the six pathogens responsible for ~73% of attributable AMR deaths, integrating the Vivli AMR Register (17.5 million harmonised isolate-drug records), Global AMR R&D Hub data, and GRAM 2019 attributable burden estimates. Base-case misalignment is 25.3% (funding share vs deaths), robust across six sensitivity variants. A Bayesian random-walk projection extends the index to 2030, widening it to 27.5% (80% CI 23.0–32.0%), driven by rising *K. pneumoniae* XDR. Pipeline cross-referencing reveals *A. baumannii* (4 late-stage projects globally) and *K. pneumoniae* are doubly neglected: underfunded **and** under-pipelined. The framework supports evidence-based reallocation toward the WHO Priority 1 carbapenem-resistant Gram-negative pathogens.

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## 1. Objectives

AMR is projected to become a leading global cause of death by 2050<sup>1</sup>. Mobilising R&D against AMR has been a sustained policy priority for a decade, with the Global AMR R&D Hub<sup>9</sup> now tracking nearly USD 19 billion in committed investment across 89 funders. A central but under-examined question is whether this investment is **proportionate to where the burden actually concentrates** at the level of individual pathogens. Existing analyses describe the pipeline<sup>4</sup> or the burden<sup>1</sup> in isolation, but do not formally measure misalignment between the two.

This work asks: **for the leading AMR pathogens, how closely does R&D funding share track attributable burden share, where is the gap compounded by a thin late-stage pipeline, and how is the misalignment likely to evolve to 2030?** We answer with a Global Misalignment Index (GMI), a pipeline cross-flag that distinguishes neglected pathogens from those served despite low funding, and a Bayesian projection of XDR prevalence that propagates uncertainty into a projected 2030 GMI.

## 2. Methods

For the surveillance layer, we unified three Vivli AMR Register<sup>8</sup> datasets (ATLAS, KEYSTONE, SIDERO-WT) into a long-format table of 17,477,785 isolate-drug records spanning 2004 to 2024 across 89 countries. Censored MICs are preserved with explicit flags; isolate identity is guaranteed unique before pivoting. Each MIC was classified S/I/R against EUCAST Clinical Breakpoint Tables v14.0<sup>2</sup>, verified line-by-line against the source PDF. Classification is censoring-aware ( $\leq v$  returns S only if  $v \leq S_{\text{max}}$ ; else Unresolved). CLSI was dropped as values could not be verified to primary source.

For resistance phenotyping, we fitted Latent Class Analysis models on *K. pneumoniae* and *A. baumannii* using `stepmix`<sup>7</sup> with binary encoding and missingness handled by full-information ML. *A. baumannii* resolved into 4 phenotypes (BIC-optimal), including an XDR<sup>5</sup> class (52%) resistant to all panel drugs except colistin. *K. pneumoniae* was fixed at K=5 by clinical parsimony (BIC favoured K=6 but the sixth was a 0.6% variant): susceptible, aminoglycoside-S ESBL, aminoglycoside-R ESBL, XDR (13%, with 19% colistin resistance), and a small carbapenem-intermediate class.

For funding attribution, we built a Funding Allocation Profile by mapping projects to pathogens via the Hub Categories coded field (e.g. 1504 Infectious Agent / Bacteria / Gram negative / *Klebsiella* spp.). Multi-pathogen projects split USD equally among named target genera; funding is recorded at project start year. Two sensitivity variants redistribute the broad-spectrum pool (projects tagged “Gram negative” or “Gram positive” without a named genus): a *proportional* variant splits it by base shares within

Gram class, and an *equal* variant splits it evenly across that class’s genera. Gram-negative and Gram-positive pools are kept separate.

Per-pathogen attributable deaths and DALYs for 2019 were extracted from the Lancet GRAM appendix<sup>1</sup>, Table S22, row “Resistance to one or more antibiotics” (attributable columns; not associated). The six pathogens sum to 929,000 attributable deaths, matching the GRAM headline. Per-pathogen misalignment is funding share minus burden share (percentage points); the Global Misalignment Index is the half-sum of absolute misalignments — the total-variation distance between the two share distributions, interpretable as the share of funding that would need to shift for perfect alignment. Three funding cases × two burden metrics yield six variants. We further count late-stage therapeutic projects (Hub Categories containing “Therapeutics / Development” or “Therapeutics / Approval & Post approval”, per WHO/PEW pipeline conventions) per pathogen, and classify pathogens into four quadrants by crossing misalignment sign with pipeline share (split at the median).

For the temporal extension, we fit a Beta-binomial random-walk model on the logit of XDR prevalence in Stan<sup>6</sup> to *K. pneumoniae* and *A. baumannii*, using observations through 2024 (2025 excluded as incomplete). The XDR class is identified programmatically by profile, not class number. The model —  $\theta_t = \theta_{t-1} + \text{Normal}(0, \sigma)$ , weakly informative priors — propagates uncertainty into a 2030 forecast. Projected prevalences translate into projected burdens under the stated assumption that attributable burden scales linearly with XDR prevalence relative to 2019. This is a strong assumption: XDR is the tail of the resistance spectrum, so scaling by the XDR ratio over-attributes burden change to that class; a carbapenem-R scaling would be more conservative but is not run. The four unprojected pathogens hold their 2019 burdens; funding shares are held constant. Uncertainty propagates to the projected GMI by recomputing from 8,000 posterior draws. Funding and burden are both at global pathogen level, neither drug-class resolved; funding is cumulative, burden is the 2019 cross-section — we frame the comparison as cumulative R&D effort against the burden it is intended to address.

### 3. Results

The Global Misalignment Index is **25.3%** for the six leading pathogens, base case (deaths). This implies that approximately a quarter of pathogen-attributed funding (USD 4.17 billion across the six in our base FAP) would need to redistribute to align with burden share. The index is stable across alternative metrics and funding assumptions (Table 1), ranging from 15.7% to 25.3% across all six variants; deaths versus DALYs barely moves the index.

**Table 1. GMI under sensitivity variants (% misalignment).**

| Funding case                           | Deaths | DALYs |
|----------------------------------------|--------|-------|
| Base (pathogen-named only)             | 25.3   | 23.0  |
| Proportional broad-pool redistribution | 19.1   | 20.2  |
| Equal broad-pool redistribution        | 15.7   | 16.7  |

The per-pathogen pattern (Figure 1) is clinically coherent. The three underfunded pathogens are Gram-negatives: *K. pneumoniae* (8% of funding against 21% of burden; gap of −12.4 percentage points), *E. coli* (−6.8 pp), and *A. baumannii* (−6.1 pp). The three overfunded pathogens are *P. aeruginosa* (+11.7 pp), *S. aureus* (+7.6 pp), and *S. pneumoniae* (+5.9 pp).

**The pipeline cross-flag sharpens this into an actionable map (Figure 1, Table 2).** *A. baumannii* and *K. pneumoniae* both fall into the **NEGLECTED** quadrant: underfunded relative to burden **and** under-represented in the late-stage pipeline. *A. baumannii* is the starkest case, with only 4 late-stage projects worldwide (5.3% of the 251 late-stage projects across our six) against 14% of attributable AMR deaths. *E. coli* falls into **SERVED DESPITE LOW FUNDING**: funding lags burden, but the late-stage pipeline (25%) is healthy, consistent with broad-spectrum and UTI-targeted development. *S. pneumoniae*’s apparent **INVESTMENT-TRANSLATION GAP** position (overfunded, pipeline 2.6%) is largely a genus-attribution

artefact: Streptococcus spp. Hub funding pools *S. pyogenes*, group B Streptococcus, and pneumococcal conjugate-vaccine work not targeting antibiotic development. We flag this quadrant as not actionable at current attribution resolution. *S. aureus* and *P. aeruginosa* are **WELL-RESOURCED**.

**Table 2. GMI with pipeline cross-flag (base funding, attributable deaths 2019).**

| Pathogen             | Funding % | Burden % | f-b (pp) | Pipeline % | Quadrant                   |
|----------------------|-----------|----------|----------|------------|----------------------------|
| <i>K. pneumoniae</i> | 8.4       | 20.8     | −12.4    | 11.8       | NEGLECTED                  |
| <i>E. coli</i>       | 16.8      | 23.6     | −6.8     | 25.0       | SERVED                     |
| <i>A. baumannii</i>  | 8.1       | 14.2     | −6.1     | 5.3        | DESPIITE LOW FUNDING       |
| <i>S. pneumoniae</i> | 19.1      | 13.1     | +5.9     | 2.6        | INVESTMENT-TRANSLATION GAP |
| <i>S. aureus</i>     | 26.8      | 19.2     | +7.6     | 30.3       | WELL-RESOURCED             |
| <i>P. aeruginosa</i> | 20.9      | 9.1      | +11.7    | 25.0       | WELL-RESOURCED             |

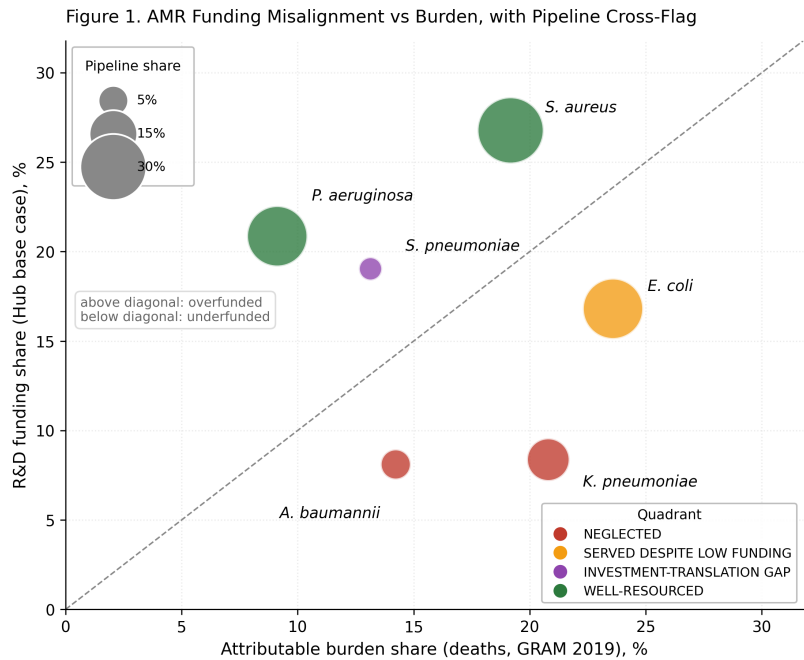


Figure 1: AMR funding misalignment vs burden, with pipeline cross-flag. Bubble size encodes late-stage pipeline share; colour encodes quadrant.

### 3.1 Projected 2030 misalignment

We fit the model separately to *K. pneumoniae* and *A. baumannii* XDR series. MCMC diagnostics were clean ( $R\text{-hat} \leq 1.01$ , satisfactory ESS; *Klebsiella* 6/8,000 divergent transitions, 0.07%). Posterior RW step SD on the logit scale was 0.30 (*A. baumannii*) and 0.17 (*K. pneumoniae*).

*A. baumannii* XDR has plateaued at ~63–65% since 2018; the 2030 posterior mean is 61.5% (80% UI 39.8–81.4%). *K. pneumoniae* XDR rose from 6.5% in 2012 to 18.6% in 2024, with 2030 posterior mean 19.4%

(80% UI 11.9–27.7%). The random walk is conservative by design — it forecasts continued movement at recent rates without extrapolating a trend — which we report as a feature against over-extrapolation.

Propagating uncertainty through the index yields a **projected 2030 GMI of 27.5% (80% CI 23.0–32.0%, 95% CI 20.7–35.1%)**, a modest widening from 25.3%. The widening is driven by *K. pneumoniae*, whose projected misalignment grows from –12.4 to –17.3 pp (80% UI –25.5 to –9.3); *A. baumannii* narrows slightly (–6.1 to –4.5 pp) as Kp’s growing share absorbs weight. Most of the widening reflects the observed 2019 to 2024 Kp XDR growth (14.5% to 18.6%) rather than the RW’s forward projection (which adds only ~1 pp of drift to 2030). The 80% UI of the projected GMI overlaps the current value: the central estimate drifts upward but we cannot claim with high confidence the index will worsen.

Figure 2. XDR prevalence with Bayesian random-walk projection to 2030

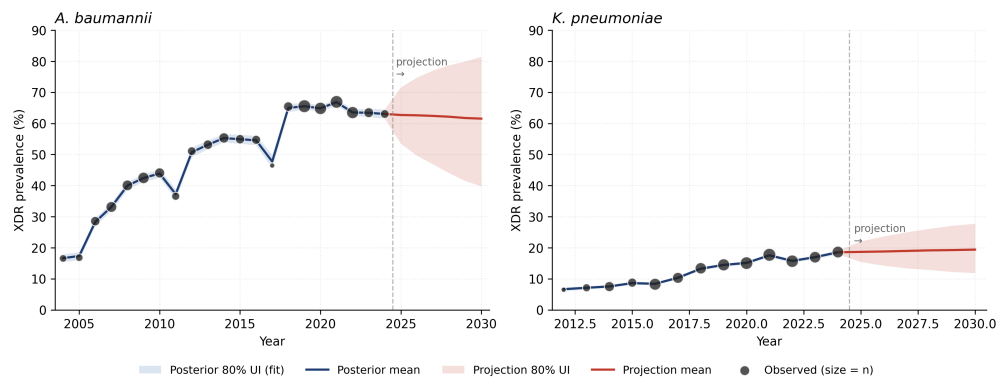


Figure 2: XDR prevalence with Bayesian random-walk projection to 2030. Observed dots sized by isolate count; blue band = posterior 80% UI over the fitted period; red fan = projection 80% UI to 2030.

## 4. Impact of the work

The doubly-neglected finding is the central message: **two of the six leading AMR pathogens (*A. baumannii* and *K. pneumoniae*) are receiving disproportionately little funding relative to the deaths they cause, and that gap is compounded by a thin late-stage pipeline.** These two pathogens are precisely the WHO Priority 1 critical-priority carbapenem-resistant Gram-negatives<sup>3</sup>. The convergence of three independent signals (burden share, funding share, pipeline share) on the same conclusion strengthens it materially. The Bayesian projection adds a fourth: the gap for *K. pneumoniae* widens further under current trends.

**Implications for R&D allocation.** Closing the *K. pneumoniae* gap (–12.4 pp) alone requires ~USD 520M redistribution against the base FAP of USD 4.17B; closing both Kp and Ab requires ~USD 770M. A partial rebalance of USD 400–600M would close half to two-thirds of the combined gap. Even the most generous redistribution assumption (equal broad-pool sensitivity) leaves a GMI of 15.7%, so the qualitative case does not depend on attribution rule. The projection suggests the case strengthens, not weakens, on a 2030 horizon.

**Limitations and caveats.** Funding is attributed at genus level (the Hub does not encode species or drug class); we use genus as a proxy for the priority species. For five of six pathogens this is a close fit; for *Streptococcus* it is the loosest, since the genus includes non-pneumoniae species. Funding is cumulative (dominated by post-2013 activity); burden is the 2019 cross-section. Project counts proxy candidate counts in the pipeline measure, biasing all pathogens similarly. The Vivli surveillance datasets carry no patient outcomes, so we rely on external GRAM attributable estimates; the SPIDAAR linkage dataset was requested but unavailable. The projection assumes attributable burden scales linearly with XDR prevalence and projects only two of six pathogens. The misalignment finding is robust to plausible variation in attribution rules and metric choice, but percentage-point precision depends on the resolution of underlying data

systems.

**Policy ask.** Funders coordinating through the Global AMR R&D Hub — including CARB-X, GARDP, Wellcome Leap, BARDA, JPIAMR, and Novo REPAIR — should institute a recurring pathogen-level alignment audit and rebalance portfolios toward the doubly-neglected pathogens identified here. The framework is reproducible (full code and decision log open), extensible, and updatable as new GRAM and Hub data are released; the Vivli AMR Register provides the surveillance granularity that makes this audit possible.

## References

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*Repository (code, decision log, reproducibility): [github.com/anthoniooladimeji11-coder/amr-funding-gap](https://github.com/anthoniooladimeji11-coder/amr-funding-gap)*

*All analyses use open data. Surveillance data accessed via the Vivli AMR Register under the 2026 Data Challenge; funding data from the Global AMR R&D Hub; burden estimates from the GRAM 2019 Lancet appendix Table S22.*